Introduction to Computer Programming 2024-25

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Switch/case statement

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Switch statement

In computer programming languages, a switch statement is a type of selection control mechanism used to alter the flow of program execution based on the value of a variable or expression.

Switch statements operate in a manner somewhat akin to the if statement and are present in most high-level programming languages, utilizing keywords like switch, case, select, or inspect.

Exercise

Write a program that given as input a programing language (Javascript, PHP, Python, Solidity, Java) print a string depending on the input.

Javascript	"You can become a web developer."
PHP	"You can become a backend developer."
Python	"You can become a Data Scientist."
Solidity	"You can become a Blockchain developer.”
Java	"You can become a mobile app developer."

Solution

```
lang = input("insert a language: ")

if lang == "Javascript":
    print("You can become a web developer.")
elif lang == "PHP":
  print("You can become a backend developer.")
elif lang == "Python":
  print("You can become a Data Scientist")
elif lang == "Solidity":
  print("You can become a Blockchain developer.")
elif lang == "Java":
  print("You can become a mobile app developer")
else:
  print(“Wrong input!”)
```

Python Switch Statement

Until version 3.10, Python never had a feature that implemented what the switch statement does in other programming languages.

So, if you wanted to execute multiple conditional statements, you would've had to use the elif keyword.

From version 3.10 upwards, Python has implemented a switch case feature called “structural pattern matching”. You can implement this feature with the match and case keywords.

Python Switch Statement

```
match term:
    case pattern-1:
         action-1
    case pattern-2:
         action-2
    case pattern-3:
         action-3
    case _:
         action-default
```

Note that the underscore symbol is what you use to define a default case for the switch statement in Python.

Lang exercise using switch/case

```
lang = input("What's the programming language you want to learn? ")

match lang:
    case "JavaScript":
      print("You can become a web developer.")
    case "Python":
        print("You can become a Data Scientist")
    case "PHP":
        print("You can become a backend developer")
    case "Solidity":
        print("You can become a Blockchain developer")
    case "Java":
        print("You can become a mobile app developer")
    case _:
        print("Wrong input!")
```

That’s a much cleaner syntax than multiple elif statements and simulating the switch statement with a function.

Elegant code

Elegant code employs cleverness to achieve a task in significantly fewer lines than one might expect, all while maintaining readability and clarity.

Above all, good code should prioritize cleanliness, simplicity, and ease of comprehension. The cleaner and simpler the code, the lower the likelihood of introducing unnoticed bugs.

Elegant code

Elegant code is a combination of:

- **Correctness**: IMHO, no incorrect code can truly be considered elegant.
- **Succinctness**: Less code implies fewer opportunities for errors and is easier to understand.
- **Readability**: Code that is easy to understand is also easier to maintain.
- **Performance**: Up to a certain point, as overly optimized code prematurely may not be considered elegant.
- **Adherence to standards**: When faced with two equally elegant options, the one that aligns with the established standard is preferred.

Exercise

Write a calculator that support the four main operations using the switch/case statement (only two inputs).

Solution

```
first = int(input("First Number: "))
second = int(input("Second Number: "))
operation = input("Insert operation +,-,/,*: " )

match (operation):
  case "+":
    print(first + second)
  case "-":
    print(first - second)
  case "/":
    if second == 0:
      print("wrong input!")
    else:
      print(first / second)
  case "*":
    print(first * second)
  case _:
    print("operation not recognized")
```

Exercise

Write a program that given a date prints past, present or future depending on the actual date.

Solution

```
import datetime

day = int(input("day: "))
month = int(input("month: "))
year = int(input("year: "))
date = datetime.datetime(year, month, day).strftime("%Y-%m-%d")
now = datetime.datetime.now().strftime("%Y-%m-%d")

if date < now:
    print('past')
elif date > now:
    print('future')
else:
    print('present')
```

Exercise

Write a program that takes a student's grade as input and print a letter grade based on the following scale:

- 90-100 A
- 80-89 B
- 70-79 C
- 60-69 D
- < 60 F

Solution

```python
# Get the student's grade as input
grade = float(input("Enter the student's grade: "))

# Determine the letter grade
if grade >= 90 and grade <= 100:
    letter_grade = 'A'
elif grade >= 80 and grade < 90:
    letter_grade = 'B'
elif grade >= 70 and grade < 80:
    letter_grade = 'C'
elif grade >= 60 and grade < 70:
    letter_grade = 'D'
elif grade < 60:
    letter_grade = 'F'
else:
    letter_grade = 'Invalid grade'

# Print the letter grade
print(f"The letter grade is: {letter_grade}")
```

Exercise

Convert the script using the `match` structure

```
name = input('Enter your name: ')
if name == 'Nic':
  # Do stuff
elif name == 'John':
  # Do stuff
elif name == 'Sofia':
  # Do stuff
elif name == 'Hans':
  # Do stuff
```

Solution

```python
name = input('Enter your name: ')

match name:
  case 'Nic':
    # Do stuff
  case 'John':
    # Do stuff
  case 'Sofia':
    # Do stuff
  case 'Hans':
    # Do stuff
```

Exercise

Write a progran that asks the user to input a fruit name and prints a description of the fruit given:

- Apple A sweet and juicy fruit that comes in many different colors and varieties.
- Banana A soft and sweet fruit that is often eaten fresh, baked, or fried.
- Orange A citrus fruit with a juicy and sour flesh.
- Grape A small, round fruit that grows in clusters. Grapes can be red, green, or black.
- Watermelon A large, round fruit with a sweet and refreshing flesh.

If the fruit is unknown it should write “I don't know what that fruit is.”

Solution

```
# Get fruit input from the user
fruit = input("Enter a fruit name: ").lower()

# Use match statement to check the fruit and print the description
match fruit:
    case "apple":
        print("A sweet and juicy fruit that comes in many different colors and varieties.")
    case "banana":
        print("A soft and sweet fruit that is often eaten fresh, baked, or fried.")
    case "orange":
        print("A citrus fruit with a juicy and sour flesh.")
    case "grape":
        print("A small, round fruit that grows in clusters. Grapes can be red, green, or black.")
    case "watermelon":
        print("A large, round fruit with a sweet and refreshing flesh.")
    case _:
        print("I don't know what that fruit is.")
```